

Optic Neuropathies 1



Imaging of the optic nerve: technological advances and future prospects

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Over the past decade, ocular imaging strategies have greatly advanced the diagnosis and follow-up of patients with optic neuropathies. Developments in optic nerve imaging have specifically improved the care of patients with papilloedema and idiopathic intracranial hypertension, inflammatory optic neuropathies, and compressive optic neuropathies. For example, optic nerve imaging with optical coherence tomography (OCT) is now widely used as an outcome measure in clinical trials of neurological disorders (eg, demyelinating diseases), and OCT findings could be informative of disease progression in patients with various neurodegenerative disorders (eg, Alzheimer's disease or Parkinson's disease). In the past 5 years, multimodality optic nerve imaging has expanded to systematically include focused and wide-field colour and autofluorescence fundus photographs; various types of optic nerve, macular, and vascular OCT; and specific MRI techniques. Such multimodality imaging makes the diagnosis of optic neuropathies easier and provides objective information on optic nerve damage, which is useful for prognosis. Non-mydratic ocular fundus cameras and OCT have become readily available in non-ophthalmic settings and could easily be implemented in neurological clinics and emergency departments, allowing for direct access to optic nerve imaging and enabling teleconsultations. In the future, these imaging studies could be used in association with artificial intelligence deep-learning systems, which are already transforming the field of ocular imaging.

Introduction

In the past decade, advances in non-invasive ocular imaging and neuroimaging techniques, such as non-mydratic ocular fundus photography and optical coherence tomography (OCT), have transformed the clinical approach to the patient with visual loss and presumed optic neuropathy. Multimodality imaging of the optic nerve is now standard in clinical practice, with each technique used providing specific information that, in combination, can facilitate diagnosis and monitoring of various optic neuropathies and neurological disorders. The ability to directly image the retinal ganglion cells, optic nerve head, and orbital optic nerve provides valuable information about the anatomy of the optic nerve. These data not only facilitate a definite diagnosis of optic neuropathy but also might indicate the specific mechanism or disorder causing the optic neuropathy.^{1–6} Furthermore, anatomical features correlate closely with functional outcomes.^{2,4}

Ocular imaging modalities have come into widespread clinical use for the diagnosis and follow-up of patients with optic neuropathies. For example, OCT is now part of the standard clinical examination in most ophthalmic practices. This technique is also in common use for monitoring of several neurological disorders, such as multiple sclerosis and other demyelinating disorders, and OCT is frequently included to measure axonal loss as an outcome measure in neurological clinical trials.⁴ Similarly, high-quality orbital imaging by MRI, CT, and PET provides important findings relevant to the diagnosis of optic neuropathies.⁶ For neurologists, access to inexpensive ocular fundus photography, a technique that has largely replaced the ophthalmoscope, has improved in the past few years.

In this Series paper, we highlight the progress that has been made in the past 5 years in optic nerve imaging, and we emphasise modalities that can be routinely used by neurologists. We discuss up-to-date information on non-mydratic ocular fundus photography, which is typically the first step in optic nerve examination; orbital ultrasound, which is part of multimodality imaging; and OCT, which has greatly improved over the past decade and provides details of optic nerve structure that are not seen on photography or ultrasound. Additionally, we briefly discuss how artificial intelligence deep-learning systems (AI-DLS), might be able to automatically recognise retinal and optic nerve abnormalities on ocular imaging. Finally, advances in the diagnosis of optic neuropathies with orbital CT and MRI are reviewed. This paper is the first in a three-part Series in *The Lancet Neurology* on optic neuropathies.^{2,3}

Diagnosis of optic neuropathies

The diagnosis of an optic neuropathy is made after obtaining a thorough patient history and doing an ophthalmological examination.¹ Clinical findings—including a decrease in central visual acuity, abnormal colour vision, visual field defects (eg, central or cecentral scotomas, or arcuate defects), presence of a relative afferent pupillary defect in unilateral or asymmetric optic neuropathies, and abnormal optic nerve head appearance—usually allow for accurate localisation of the lesion to the optic nerve and a presumptive differential diagnosis of the underlying pathophysiology (appendix p 2).^{1–3} Use of multimodality ocular and orbital imaging—eg, with ocular fundus photography, ultrasound, OCT, or orbital CT or MRI—supports clinical findings and

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See Online for appendix

	Retinal cause of visual loss (maculopathy)	Optic neuritis (inflammatory)	Anterior ischaemic optic neuropathy	Compressive or infiltrative optic neuropathy	Leber hereditary optic neuropathy	Toxic, nutritional optic neuropathy	Papilloedema* versus pseudo-disc oedema	Glaucoma
Ocular fundus photography								
Early changes	Might show macular changes, but can be subtle and missed	Optic nerve normal or swollen; look for neuroretinitis	Optic nerve swollen; fellow eye with small cup-to-disc ratio†	Optic nerve normal or swollen (if lesion in anterior orbit) or pale if insidious onset	Optic nerve normal or hyperaemic with telangiectasia (pseudo-disc oedema)	Bilateral optic nerve normal or pale if insidious onset	Bilateral optic nerve swelling, or optic nerve elevation, drusen; autofluorescence reveals optic nerve drusen	Localised loss of the pRNFL bundle; localised notch of the neuroretinal rim
Chronic findings	Macular changes can be subtle; attenuated retinal arteries; optic nerve pallor might develop	Swelling resolves; optic nerve pallor develops	Swelling resolves; optic nerve pallor develops	Optic nerve pallor; optic nerve cupping†	Optic nerve pallor is delayed by a few months; optic nerve cupping†	Bilateral optic nerve pallor	Bilateral optic nerve swelling or secondary optic atrophy if true papilloedema; no change if pseudo-disc oedema	Bilateral optic nerve vertical cupping (can be asymmetric); cupping more than pallor; peripapillary atrophy
Follow-up signs	Progression of macular changes	Optic nerve pallor	Optic nerve pallor; no swelling	Optic nerve pallor; optic nerve cupping†	Optic nerve pallor; optic nerve cupping†	Bilateral optic nerve pallor	Changes in oedema if swollen (optic nerve flattens or remains elevated, or with secondary optic atrophy)	Increasing cupping and peripapillary atrophy
Ocular ultrasound								
Observed changes	Abnormal if there is mass or retinal detachment; most often there are NOCs	Optic nerve might be elevated if major swelling; optic nerve might be thickened	Optic nerve might be elevated if major swelling	NOCs; anterior orbital mass might be seen	NOCs	NOCs	Optic nerve might be elevated if major swelling; optic nerve sheath might be dilated; calcified optic nerve drusen seen	NOCs or cupping
OCT, pRNFL, and mGCIPL protocols								
Early changes	Abnormal macula (all layers)	pRNFL normal or thick (if optic nerve swollen initially); mGCIPL normal unless neuroretinitis	pRNFL thick (optic nerve swollen); mGCIPL normal	pRNFL thick if optic nerve swollen, normal‡ if acute, or thin if chronic; mGCIPL normal or thin if chronic visual loss	pRNFL mildly thick; mGCIPL normal	Bilateral pRNFL normal if acute or thin if chronic; mGCIPL normal initially or thin if chronic visual loss	pRNFL thick if optic nerve swollen; mGCIPL normal; Bruch's membrane angled if intracranial hypertension; EDI-OCT might show optic nerve drusen or PHOMS‡	Localised thinning of the pRNFL superiorly or inferiorly; localised temporal thinning of the mGCIPL
Chronic findings	Abnormal macula (all layers); mGCIPL might become thin	mGCIPL becomes thin first, then pRNFL becomes thin	pRNFL thickness normalises and mGCIPL becomes thin first, then pRNFL becomes thin	mGCIPL becomes thin first, then pRNFL thin	mGCIPL becomes thin first, then pRNFL thin	mGCIPL becomes thin first, then pRNFL thin	pRNFL thickness normalises as optic nerve swelling resolves; mGCIPL thin if damage (optic atrophy)	pRNFL thinning of the superior or inferior quadrants, or both; mGCIPL thinning sparing the PMB
Follow-up signs	Progression of macular changes	pRNFL and mGCIPL thinning worsen	pRNFL and mGCIPL thinning stable	pRNFL and mGCIPL thinning worsen with compression	pRNFL and mGCIPL thinning stable	pRNFL and mGCIPL thinning worsen or stable	pRNFL thicker if papilloedema or thinner if damage; mGCIPL thinning if damage ongoing	Progression of the pRNFL to other quadrants; diffuse mGCIPL thinning

(Table 1 continues on next page)

often provides insight into the cause of the optic neuropathy (table 1). Remote assessment of patients, through telemedicine, is already possible (panel 1).

Ocular fundus photography

Since digital non-mydriatic cameras were introduced into routine ophthalmic practice in the 1990s, ophthalmoscopy has progressively been complemented by ocular fundus photography (appendix p 3).^{17–22} Cameras developed in the past decade provide very sharp

high-resolution images, with pupil tracking and autofocus options, wide-field options, and wireless secure remote transfer of photographs (panel 2).^{9,21} High-quality handheld non-mydriatic fundus cameras, and even smartphone photography, are attractive alternatives to ophthalmoscopy and might enable routine fundus photography by non-ophthalmologically trained providers in non-ophthalmic settings,^{9,18,21} including for children as young as 2 years.^{23,24} Additionally, several studies have highlighted the preferred use of fundus

	Retinal cause of visual loss (maculopathy)	Optic neuritis (inflammatory)	Anterior ischaemic optic neuropathy	Compressive or infiltrative optic neuropathy	Leber hereditary optic neuropathy	Toxic, nutritional optic neuropathy	Papilloedema* versus pseudo-disc oedema	Glaucoma
(Continued from previous page)								
Orbital CT								
Early changes	NA	Optic nerve often NOCs; optic nerve enhancement might be seen if severe diffuse optic nerve inflammation	NOCs	Optic nerve compression by orbital or intracranial mass; optic nerve infiltration or optic nerve mass	NOCs	NOCs	Swollen optic nerve head might appear elevated; signs of intracranial hypertension§; calcified optic nerve head drusen might be seen	NOCs
Orbital MRI¶								
Early changes	NA	Optic nerve enhancement; optic nerve sheath and orbital enhancement suggest MOGs or systemic disease; brain MRI might be atypical if demyelinating disease	NOCs (no orbital optic nerve enhancement), although swollen optic nerve head might appear elevated. MRI-DWI usually not sensitive enough to show restricted diffusion on optic nerve	Optic nerve compression by orbital or intracranial mass; optic nerve infiltration or optic nerve mass	Normal optic nerve with no enhancement (usual); faint enhancement of mildly enlarged intracranial optic nerve and chiasm might rarely be seen	Normal optic nerve with no enhancement	Swollen optic nerve head might appear elevated; signs of intracranial hypertension‡	NOCs
Follow-up signs	NA	Optic nerve enhancement most often resolves; optic nerve atrophy might develop with hyper T2 signal; optic nerve enhancement if recurrent inflammation	Normal optic nerve; optic nerve atrophy might develop with hyper T2 signal if severe visual loss	For mass recurrence or growth; optic nerve atrophy might develop with hyper T2 signal	Normal optic nerve; optic nerve atrophy might develop with hyper T2 signal	Normal optic nerve; optic nerve atrophy might develop with hyper T2 signal	Normal optic nerve; signs of intracranial hypertension often do not resolve once intracranial hypertension is treated	Atrophy of the optic nerve or optic chiasm, or both; possible hyper T2 signal of optic nerve
Early findings include those changes observed from onset of visual loss to a few weeks after visual loss (usually before 6 weeks). Chronic findings include those observed more than 6 weeks after visual loss onset. DWI=diffusion-weighted imaging. EDI=enhanced depth imaging. mGCIPL=macular ganglion cell-inner plexiform layer. MOGs=myelin-oligodendrocyte glycoprotein antibodies. NA=not applicable. NOCs=no obvious changes. OCT=optical coherence tomography. PHOMS=peripapillary hyper-reflective ovoid mass-like structures. PMB=papillomacular bundle. pRNFL=peripapillary retinal nerve fibre layer. *The term papilloedema specifically indicates optic nerve head swelling secondary to intracranial hypertension. †The cup-to-disc ratio measures the size of the central cup in the optic nerve head (diameter of the cup: diameter of the optic disc) visible on funduscopic examination. The physiological central cup ranges from 0.0 to 0.8. A physiological small cup (<0.2) is a risk factor for non-arteritic anterior ischaemic optic neuropathy. Patients with a large cup-to-disc ratio (>0.6) might have a physiological enlarged cup (normal), or might have an optic neuropathy typically associated with progressive cupping of the optic nerve head such as glaucomatous optic neuropathies, some compressive optic neuropathies, hereditary optic neuropathies, or after acute arteritic (from giant cell arteritis) anterior ischaemic optic neuropathy. Optic nerve cupping is seen on ocular fundus photography and OCT. Optic atrophy refers to the colour of the optic nerve rim (pink or orange in normal optic nerves and pale in atrophic nerves), whereas optic nerve cupping refers only to the size of the cup regardless of the colour of the rim. Glaucomatous optic neuropathies have a large cup with a pink rim whereas other cupping optic neuropathies have a pale rim with a large cup. ‡Common but non-specific OCT marker of axoplasmic stasis in the optic nerve head, often found after chronic papilloedema has resolved. §Radiological signs of intracranial hypertension include optic nerve head protrusion often seen as optic nerve head enhancement (corresponding to the swollen optic nerve head), optic nerve sheath dilation in the orbit, vertical optic nerve tortuosity, partly empty sella, cerebellar tonsillar ectopia, enlarged Meckel's cave, encephaloceles and meningoencephaloceles, skull base erosion, and bilateral transverse venous sinus stenosis. ¶Orbital MRI and CT must be performed both without and with contrast (with fat suppression for MRI) to allow for optic nerve evaluation.								
Table 1: Summary of optic nerve imaging for the diagnosis and follow-up of common optic neuropathies								

photography over funduscopy to teach medical students the art of viewing the ocular fundus.^{19,22,25} By removing the technical challenges of direct ophthalmoscopy, examination of the ocular fundus via photographs is easier and more effective, providing additional time (previously spent learning how to use the direct ophthalmoscope) for learners to appreciate key ophthalmological signs and pathology (figure 1).

Neurologists have similarly adopted non-mydratic fundus photography as an alternative to direct ophthalmoscopy.^{18,19} Examination of the optic nerve provides essential information in many neurological disorders, including headaches, intracranial hypertension, stroke, and demyelinating diseases. Findings of papilloedema and other optic neuropathies usually affect

diagnostic strategies by providing clues to the underlying disorder, and they might change the management of numerous neurological conditions to prevent irreversible visual loss.¹⁸ Reports have highlighted the use of non-mydratic fundus photography for the diagnosis of optic disc oedema by neurologists.^{26,27} Because many neurological disorders associated with abnormal optic nerve appearance are acute, use of fundus photography was validated in the emergency department for patients with headaches and acute neurological symptoms²⁸ and for those in intensive care units with subarachnoid haemorrhage.²⁹ As shown in the prospective Fundus photography versus Ophthalmology Trial Outcomes in the Emergency Department (FOTO-ED) study,³⁰ which compared direct ophthalmoscopy with non-mydratic

Panel 1: Telemedicine and optic nerve imaging

Remote stroke and neurological care using telemedicine has become popular in the past 5 years, with many neurologists providing consultations and making acute treatment recommendations during virtual visits.⁷ Strategies to facilitate remote clinical examination and interpretation of tests have been developed, including for transfer of brain and ocular imaging studies.⁷ Widespread access to ocular fundus cameras has also facilitated the expansion of teleophthalmology.⁸ For example, diabetic retinopathy screening programmes have become standard of care across health-care systems worldwide.⁹ Other teleophthalmology screening programmes that use retinal digital fundus photography can detect many ocular disorders, ranging from retinopathy of prematurity to glaucoma.^{8,10} Ocular fundus cameras are routinely used in emergency departments as part of telemedicine programmes,¹¹ and are especially helpful in triaging patients with visual complaints or those with suspected intracranial hypertension when no eyecare provider is readily available.^{8,10} The COVID-19 pandemic prompted the rapid expansion of teleophthalmology programmes, including for neuro-ophthalmic evaluations, with widespread implementation of non-mydratic fundus cameras.^{11–14} Additionally, the fear of SARS-CoV-2 infection related to close face-to-face contact with patients pushed most educators and providers to abandon the use of direct ophthalmoscopes and replace them with safer devices capable of providing images of the ocular fundus at arms-length from the patient.¹⁵ It would not be surprising if the COVID-19 pandemic has precipitated the complete abandonment of the direct ophthalmoscope in most non-ophthalmic settings.¹⁶

fundus photography, emergency department providers rarely use the ophthalmoscope, and non-mydratic fundus photography is an effective alternative to examination of the ocular fundus in the emergency department.³¹ Similar results were shown in Australian emergency departments³² and in an outpatient neurology clinic.³³ Despite these advances, and the rapidly decreasing cost of high-quality non-mydratic fundus cameras, many neurologists and emergency department providers continue to ignore the value of the ocular fundus examination and do not attempt to examine the optic nerves as part of their routine evaluations.^{18,34} Resulting diagnostic errors can culminate in devastating outcomes—eg, overlooking papilloedema in a patient with headaches.³⁵ Low confidence in non-ophthalmology trained health-care providers in interpreting ocular fundus photographs, without the help of an eyecare provider, could be a reason for the underuse of this readily available technology.³⁴ However, remote interpretation of these images by an eyecare provider, or by use of AI-DLS (panel 3), could potentially solve this issue.

Ocular ultrasound

Ocular ultrasound is routinely used in ophthalmology clinics to evaluate all parts of the eye, including the anterior optic nerve. This technique can be used to visualise optic nerve elevation and optic nerve tumours, and it is especially useful to identify calcified optic nerve head drusen as hyperechogenic foci at low gain.^{42,43} In the past decade, the role of ocular point-of-care ultrasound (POCUS) has been emphasised in patients with ophthalmic emergencies, particularly in the emergency department or in intensive care units where ophthalmic health-care providers might not be readily available.^{43–45} Although POCUS is most useful to evaluate ocular trauma and diagnose vitreous haemorrhage and retinal detachment, some reports have suggested that it could be used to identify optic nerve swelling, suggestive of intracranial hypertension in the correct clinical setting.^{44,45} POCUS might also show optic nerve thickening in patients with suspected optic neuritis.⁴⁶ The technique is widely available, easy to perform, and can provide information in emergency settings when funduscopy is difficult and fundus cameras are not available.

Optical coherence tomography

Spectral domain OCT is an increasingly available non-invasive imaging modality that uses infrared light reflection to capture cross-sectional, micrometre-scale, high-resolution images of the retina and optic nerve head.⁴⁵ The neuronal and unmyelinated axonal layers within the retina can be distinguished according to different intensities of reflected signal strength. Automated segmentation is also available to allow for quantitative measurement of the thickness and volume of each layer.⁴ In the neuro-ophthalmic setting, the peripapillary retinal nerve fibre layer (pRNFL) and the macular ganglion cell layer (mGCL) are of particular interest, representing the retinal ganglion axonal fibres and cell bodies, respectively, which eventually synapse within the lateral geniculate nucleus.⁴ The pRNFL is typically assessed by a ring scan around the optic nerve head (appendix p 7). Because the mGCL can be difficult to distinguish from the underlying inner plexiform layer, these layers can be collectively segmented (macular ganglion cell-inner plexiform layer [mGCIPL]) in some algorithms (appendix p 15).⁴ Advances from the past 5 years also include development of OCT angiography (OCT-A), a technique that permits high-resolution imaging of the retinal vasculature without the injection of fluorescein dye, which is necessary for retinal fluorescein and indocyanine green angiography (appendix pp 26–28).⁵

The pRNFL will show thickening with any acute anterior optic neuropathy that entails swelling. As damage occurs and optic nerve fibres undergo retrograde or antegrade dropout, thinning of the mGCIPL becomes apparent at about 3–4 weeks after optic nerve injury. Reduction of the thickness of the pRNFL might

Panel 2: Types of ocular fundus cameras

General comments on cameras

- Numerous types and brands of digital ocular fundus cameras exist around the world
- The quality of images and ease of use vary greatly depending on the camera
- Some cameras have an integrated liquid-crystal display screen—some very small and some as large as a tablet—allowing visualisation of the photograph in real time, whereas others require connection to a computer
- Some cameras allow for imaging of the retina as well as the anterior segment of the eye
- Some cameras allow only still photographs whereas others also allow for video acquisition
- All cameras take fundus photographs of one eye at a time and the best cameras have automated functions such as pupil tracking and autofocus, which greatly facilitate the process and ensure high-quality images
- All cameras use a flash, which might be uncomfortable for some patients
- Most currently available cameras allow non-mydriatic fundus photography, which means that good quality photographs can be obtained without pharmacological dilation of the pupils, saving time and greatly facilitating rapid fundus photography, with most pictures taking no more than a few minutes
- To ensure good quality, it is better to place the patient and the camera in a dark room or to have a hood over the patient's head and camera
- Non-mydriatic pictures might be of poor quality in older patients with cataracts or small pupils

Table-top cameras

- Advantages of table-top cameras are that they have the most functions, provide excellent quality non-mydriatic fundus images, and are the easiest to use (usually completely automated)
- Disadvantages are that they are the bulkiest cameras and the most expensive (particularly the hybrid cameras)
- Additional benefits of these cameras are that photographs can be taken of the peripheral retina, and montage photographs and stereo pictures are possible; autofluorescence imaging is available on some cameras, and some hybrid cameras combine colour photography, autofluorescence, and optical coherence tomography

- The field of view with table-top cameras is most often 30° or 45°, which is excellent for the posterior pole of the eyes; photographs of peripheral retina and montage photographs are possible for most cameras; some cameras have a wide field, but are not good for optic nerve imaging
- Some table-top cameras allow wireless transfer of images, but most require connection to a computer; direct connection to electronic medical records is possible for some cameras, and storage capacity is large

Handheld cameras

- Advantages of handheld cameras are that they are cheaper than table-top cameras, are portable, and allow fundus photography at the bedside
- Disadvantages are that they are more difficult to use than are table-top cameras (eg, with respect to alignment of light and obtaining non-mydriatic images), need recharging, are fragile, and produce lower-quality photographs than table-top cameras
- Many types are available (eg, off-the-shelf point-and-shoot cameras and integrated adaptor-based cameras), and prices varying greatly depending on the technology
- The field of view with handheld cameras is usually 30°, 40°, or 45°, which is excellent for the posterior pole of the eyes; photographs of peripheral retina and montage photographs are possible for some cameras
- Some cameras allow wireless transfer of images, but most require connection to computer, and storage capacity is low

Smartphone-based ophthalmic cameras

- Advantages of smartphone-based cameras are that they are small and cheap, the video function is very simple to use, and wireless transfer of images is easy
- Disadvantages are their difficulty of use, the small field of view that is limited to the optic nerve or macula (most often 10–25°, and up to 45°), low storage capacity, and difficulties with secure transfer of images
- Most currently available smartphone-based devices function better with pharmacological dilation of the pupils (mydriatic), because of the difficulty with beam alignment and because the bright built-in flash of smartphones results in constriction of the pupils, rendering image capture difficult

reflect either early resolution of swelling in anterior optic neuropathies or the beginning (at about 6 weeks) of axonal loss in all types of optic neuropathies, which starts around the same timeframe as clinically observable optic disc pallor (appendix pp 12,17). The thickness values of the pRNFL and mGCIPL that are provided by the OCT machine are automatically compared with a large database of normative values based on sex and age, and can vary slightly depending

on the type of machine used. The range of normal pRNFL thickness is wide (around 85–105 μm) as is the range for normal mGCIPL thickness (around 75–100 μm). Optic atrophy will result in progressive thinning of the pRNFL and mGCIPL, to as low as around 50 μm . The axial resolution of modern OCT is about 4 μm and the inter-examination variability is also about 4 μm , with variance based on the type of OCT machine used. Detection of small changes over time in

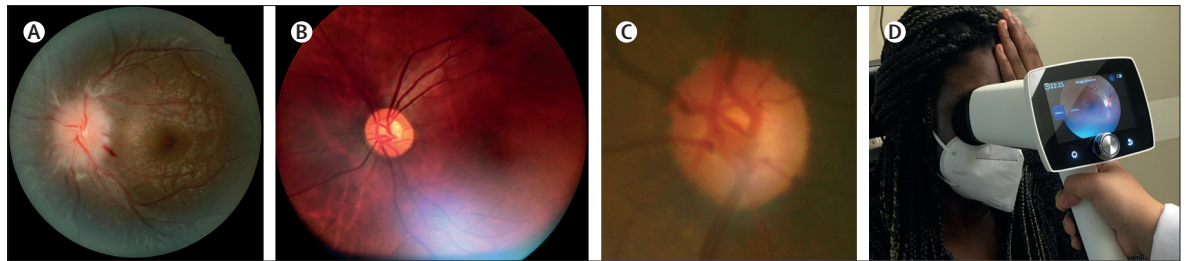


Figure 1: Examples of ocular fundus photographs obtained with non-mydratric digital fundus cameras

(A) Photograph of a swollen optic nerve obtained in the neurology clinic with a non-mydratric table-top camera (Kowa Nonmyd-alpha-D, Kowa, Japan) providing a 45° field of view. (B) Photograph of a normal optic nerve obtained in the intensive care unit with a non-mydratric handheld camera (Volk Optical Pictor Plus Fundus Camera, Volk Optical, OH, USA) providing a 40° field of view. (C) Photograph of a normal optic nerve obtained at bedside with a smartphone device (D-EYE Smartphone-Based Retinal Imaging System, D-EYE Srl, Italy); the field of view is very small without pharmacological dilation of the pupils (<10°), but it is possible to evaluate the optic nerve appearance, particularly when using the video function. (D) Handheld fundus photography without pharmacological dilation of the pupils (Optomed Aurora IQ handheld camera, Optomed, Finland) providing a 50° field of view.

Panel 3: Artificial intelligence applications of ocular imaging

AI-DLS (artificial intelligence deep-learning systems) can diagnose various ocular conditions, and numerous studies have shown that automated interpretation of ocular fundus photographs is possible.^{36,37} In 2018, the US Food and Drug Administration authorised the first autonomous use of AI-DLS for the detection of diabetic retinopathy, allowing non-ophthalmology trained personnel to detect diabetic retinopathy in primary-care settings without interpretation of photographs by an eyecare provider.³⁷ This change in the approach to the diagnosis of diabetic retinopathy augments possibilities for the applications of AI-DLS to the field of neuro-ophthalmology, suggesting that the same approach can be used for patients with neurological disorders in need of various ocular imaging studies. Indeed, AI-DLS can also be used to diagnose glaucomatous and other optic neuropathies.^{36,38} The Brain and Optic Nerve Study with Artificial Intelligence (BONSAI) study validated an AI-DLS, which could distinguish papilloedema from healthy optic nerves and recognise other optic nerve head abnormalities with 96% sensitivity and 85% specificity.³⁸ The same AI-DLS accurately and automatically graded the severity of papilloedema³⁹ and performed as well as trained neuro-ophthalmologists without clinical information.⁴⁰

Once validated in real-life situations, AI-DLS could be used as a diagnostic aid, allowing the recognition of papilloedema, normal optic nerves, and other optic nerve head abnormalities on ocular fundus photographs in emergency and neurological settings without always needing the help of an eyecare specialist on site or via teleophthalmology. The current and expected scarce access to neuro-ophthalmologists should make such tools invaluable, at least in the initial screening process, thereby helping prevent diagnostic errors.⁴¹

a patient requires the use of the same type of OCT machine at each encounter, because OCT measurements cannot be directly compared from one type of machine to another.

Use of OCT in multiple sclerosis and optic neuritis

In the past decade, growing evidence has indicated the potential diagnostic and prognostic utility of OCT parameters for patients with multiple sclerosis (appendix pp 29–33).^{2,47} Early case-control studies have consistently reported pRNFL and mGCIPL thinning, both in patients with multiple sclerosis-associated optic neuritis and in those with multiple sclerosis without optic neuritis, compared with healthy controls; however,

these anatomical changes are more pronounced among those with optic neuritis (appendix p 17).⁴⁷ In addition to pRNFL and mGCIPL thinning, a small degree of macular inner nuclear layer thickening has also been detected in patients with multiple sclerosis-associated optic neuritis compared with healthy controls, which is thought to represent inflammatory disease activity.⁴⁷ Studies from the past 2 years have suggested that a reduction in superficial macular and peripapillary vascular densities on OCT-A occurs in patients with multiple sclerosis-associated optic neuritis and in patients with multiple sclerosis who do not have optic neuritis,^{48,49} and inter-eye differences correlate well with visual function outcomes.⁵⁰ The potential role of OCT parameters in monitoring the multiple sclerosis disease course has also been highlighted by several large prospective studies and clinical trials^{2,4,47} in which a correlation was noted between pRNFL and mGCIPL measurements and clinical, biochemical, and radiological markers of multiple sclerosis (appendix p 29). Cross-sectional studies have shown an association between pRNFL and mGCIPL thinning and worse visual function in patients with multiple sclerosis, including reduced low contrast visual acuity and dyschromatopsia.^{51,52} Reduced pRNFL and mGCIPL measurements were also shown to correlate with increased intrathecal B-cell frequencies and IgG synthesis, and poorer Expanded Disability Status Scale scores.^{53,54} Changes in the pRNFL and mGCIPL were reported to be associated with various brain MRI parameters in patients with multiple sclerosis, including brain parenchymal, grey matter, white matter, caudate, and thalamus volumes, and the number and volume of cortical and optic radiation lesions.^{55–57} Moreover, mounting evidence suggests that OCT parameters might predict visual function and disability progression in patients with multiple sclerosis. For example, baseline pRNFL and mGCIPL measurements were shown to predict Expanded Disability Status Scale score progression and cognitive decline in patients with multiple sclerosis,^{58,59} whereas serial changes of pRNFL and mGCIPL within the first 1–2 months of acute optic

neuritis are associated with long-term low-contrast visual acuity, colour vision, and visual field recovery.^{60,61}

Use of OCT in other neuroinflammatory conditions

Contemporaneous studies have investigated the role of OCT in monitoring disease progression in other neuroinflammatory conditions, including neuromyelitis optica spectrum disorder (NMOSD) and myelin-oligodendrocyte glycoprotein antibody-associated disease (MOGAD). Evidence from the past 5 years suggests that pRNFL and mGCIPL thinning is more severe and more frequently bilateral following optic neuritis flares in both patients with NMOSD and MOGAD than in patients with multiple sclerosis, although no significant differences have been detected between NMOSD and MOGAD groups.^{2,62} By contrast with the strong predilection towards asymmetrical temporal sector pRNFL thinning in patients with multiple sclerosis-associated optic neuritis, optic nerve atrophy tends to be more severe and diffuse in patients with NMOSD.^{47,63} Furthermore, OCT-A studies in patients with NMOSD have reported reduced macular and peripapillary vessel densities, enlarged foveal avascular zones, and reduced foveal thickness in both patients with and without previous optic neuritis flares.⁶⁴

Use of OCT in papilloedema

The use of OCT is common for monitoring of optic nerve head oedema from intracranial hypertension,⁶⁵ and for differentiating true papilloedema from pseudopapilloedema (appendix p 20).^{42,66} The pRNFL thickness and optic nerve head volume are significantly increased in patients with papilloedema from intracranial hypertension compared with healthy controls, and several studies have also supported the positive correlation between pRNFL thickness and lumbar puncture opening pressure.⁶⁷ Decreased pRNFL measurements occur after treatment for intracranial hypertension and are a prognostic marker of long-term visual field improvement.⁶⁵ Nevertheless, caution needs to be applied when interpreting the reduction in pRNFL measurements, since the resolution of optic nerve head oedema after the treatment of intracranial hypertension can be confounded by axonal loss secondary to optic nerve atrophy.⁶⁸ Routine assessment of visual function with perimetry, therefore, remains standard of care for serial monitoring of patients with intracranial hypertension and papilloedema.^{1,68} Evaluation of mGCIPL can be especially helpful, because measurements decline with optic atrophy but do not increase with papilloedema.⁶⁸ Studies from the past decade have also highlighted that OCT has higher sensitivity than fundus photography for various retinal signs of intracranial hypertension, including peripapillary wrinkles, retinal and choroidal folds, and anterior displacement of the peripapillary retinal pigment epithelium and Bruch's membrane.^{69–71} These imaging markers worsen with intracranial hypertension, posterior scleral

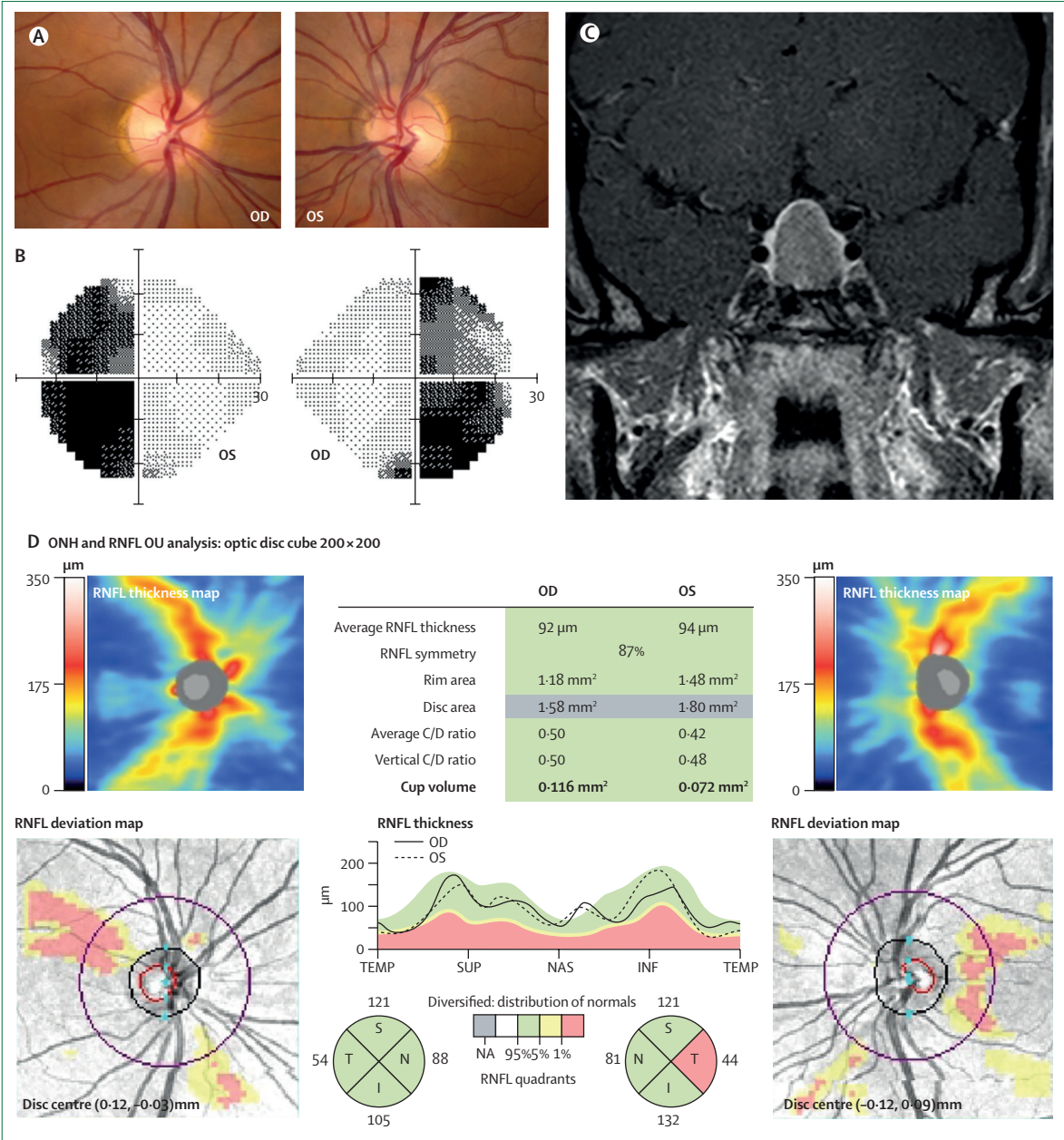
flattening, and optic nerve oedema, but resolve with pressure lowering treatment.^{69–71} Additionally, OCT with enhanced depth imaging (EDI-OCT) can also be useful for distinguishing true optic disc oedema from pseudopapilloedema secondary to optic disc drusen, which appear as hyporeflective masses with hyperreflective borders (appendix p 23).⁶⁶ Peripapillary hyperreflective ovoid mass-like structures (PHOMS) are commonly found in patients with chronic papilloedema and could represent the herniation of distended axons into the peripapillary retina. PHOMS are easily identified by EDI-OCT.⁶⁶

Use of OCT in compressive optic neuropathies

The diagnostic and prognostic performance of pRNFL and mGCIPL measurements on OCT in patients with compressive optic neuropathies is well established (figure 2).^{1,4,72–79} In chiasmal compression, preferential damage of the decussating axonal fibres originates from the nasal hemiretina, which results in binasal mGCIPL atrophy (figure 2).⁷² Cross-sectional studies have reported that pRNFL and mGCIPL measurements are thinner in patients with compressive optic neuropathies than in healthy controls. Although a strong correlation has been reported between both these OCT parameters and the severity of preoperative visual field defects,^{73,74} studies from the past 5 years have suggested that mGCIPL thinning precedes the onset of visual field changes in asymptomatic patients with chronic optic nerve or chiasmal compression.⁷⁵ Furthermore, preliminary OCT-A studies have shown that peripapillary and superficial macular vessel densities are reduced in patients with compressive optic neuropathies secondary to pituitary adenomas, compared with healthy controls.⁷⁶ Preoperative pRNFL and mGCIPL values are strong prognostic markers for both the rate of and extent of long-term postoperative visual field and visual acuity recovery.^{74,77,78} Multivariable risk prediction models incorporating age, preoperative visual function, and OCT parameters, with or without MRI compression grading, have been developed, showing moderate-to-high performance for prognosticating long-term visual field and visual acuity recovery.^{78,79}

Use of OCT in other optic neuropathies

The use of OCT can also be helpful for the diagnosis and monitoring of ischaemic, metabolic, and hereditary optic neuropathies. In non-arteritic anterior ischaemic optic neuropathy, pRNFL swelling occurs in the acute phase, followed by chronic thinning secondary to optic nerve atrophy.^{1,80,81} However, mGCIPL measurements show a stronger correlation with visual impairment in non-arteritic anterior ischaemic optic neuropathy than do pRNFL values.⁸⁰ Thinning of mGCIPL typically occurs in an altitudinal pattern corresponding to the vascular territory affected and the resultant functional visual field impairment,⁸¹ and occurs earlier than pRNFL atrophy.⁸⁰ In



(Figure 2 continues on next page)

optic neuropathies secondary to toxic or nutritional causes, such as with ethambutol toxicity or vitamin B12 deficiency, there is sectoral thinning of the temporal pRNFL and diffuse mGCIPL atrophy, which is attributed to Wallerian degeneration and the preferential effect on the papillomacular fibres.⁸² In Leber hereditary optic neuropathy, mGCIPL thinning was shown to be a sensitive marker of disease progression among carriers, and occurs before pRNFL thickening during the acute phase, followed by subsequent development of chronic pRNFL atrophy.^{3,83}

Use of OCT in neurodegenerative disorders

The potential application of OCT in neurodegenerative disorders has gained traction during the past decade (appendix pp 34–40). Studies have reported a decrease in pRNFL and mGCIPL values, and reduced macular thickness and volume, among individuals with Alzheimer’s disease compared with healthy controls. Thinning of the pRNFL is especially pronounced in the superior and inferior sectors and corresponds to the loss of magnocellular pathway retinal ganglion cells.⁸⁴ Additionally, preliminary OCT-A studies suggest that

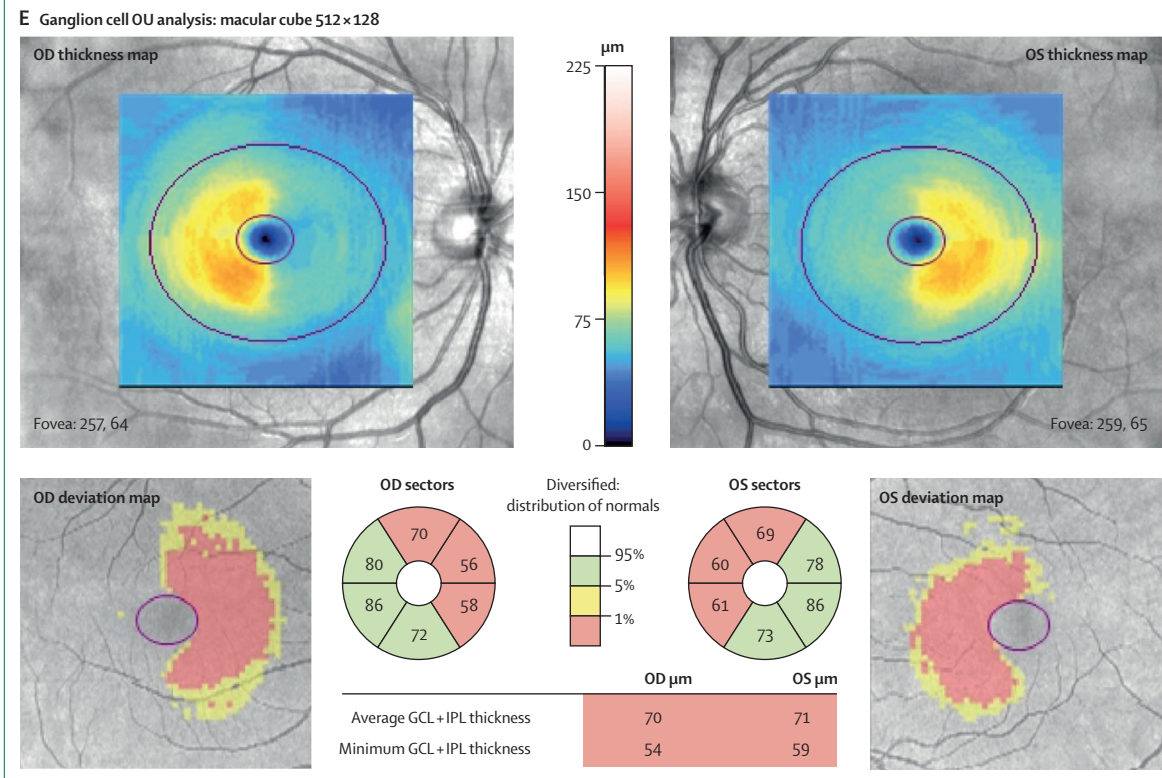


Figure 2: Example of an OCT pRNFL and mGCIPL results from compressive optic neuropathies from sellar mass

Imaging studies done at the time of discovery of a large sellar mass in a patient with an approximately 4-week history of blurry vision in both eyes. (A) Ocular fundus photographs show relatively normal-appearing optic nerves in both eyes, with perhaps mild temporal pallor. (B) Visual acuity and colour vision are normal, but 24-2 Humphrey visual fields (Humphrey Field Analyzer 3, Carl Zeiss Meditec, CA, USA) show a dense bitemporal hemianopia consistent with a chiasmal syndrome. Darker areas correspond to areas not well seen. (C) The pituitary protocol MRI (T1-weighted coronal image with contrast) shows an enhancing large sellar and suprasellar mass with compression and elevation of the optic chiasm and both intracranial optic nerves. (D) The optical coherence tomography of the pRNFL shows normal average pRNFL, measuring 92 μm in the right eye and 94 μm in the left eye. (E) However, the mGCIPL is abnormal in a pattern consistent with chiasmal syndrome. The average mGCIPL measures 70 μm in the right eye and 71 μm in the left eye and appears in red on the deviation map, corresponding to less than 1% of the thickness in the general population. The thinning is obvious in both nasal retina, corresponding exactly to the bitemporal hemianopia seen on the visual field test (the temporal visual field defect in each eye corresponds to the nasal mGCIPL loss shown in red because the nasal retina is responsible for vision in the temporal visual field; see in appendix pp 6–22). C/D=cup-to-disc ratio. GCL=ganglion cell layer. I=inferior. INF=inferior. IPL=inner plexiform layer. mGCIPL=macular ganglion cell-inner plexiform layer. N=nasal. NA=not applicable. NAS=nasal. OCT=optical coherence tomography. OD=oculus dexter. ONH=optic nerve head. OS=oculus sinister. OU=oculus uterque. pRNFL=papillary RNFL. RNFL=retinal nerve fibre layer. S=superior. SUP=superior. T=temporal. TEMP=temporal.

the foveal avascular zone is substantially enlarged, and superficial parafoveal and whole-vessel density is reduced, in people with Alzheimer's disease.⁸⁵ Cross-sectional studies have shown that pRNFL, mGCIPL, and macular thicknesses are decreased in people with preclinical Alzheimer's disease or mild cognitive impairment.⁸⁴ Moreover, a few longitudinal studies have reported that pRNFL thinning might independently predict future decline in cognitive function and development of Alzheimer's disease and dementia.⁸⁶ However, additional prospective prognostic studies are required to corroborate the predictive potential of OCT markers in disease progression.

Changes in OCT parameters have also been consistently documented in people with Parkinson's disease. Substantial thinning of the pRNFL and mGCIPL in patients with Parkinson's disease, compared with healthy controls, has been reported by many

cross-sectional studies, with preferential sectoral thinning of the inferior and temporal pRNFL.⁸⁷ Preliminary OCT-A studies have also reported decreased superficial retinal capillary plexus in people with Parkinson's disease.⁸⁸ To date, few studies have examined the correlation between OCT markers and clinical parameters of Parkinson's disease, although macular and foveal thickness measurements have been reported to be correlated with some functional measures, including the Unified Parkinson's Disease Rating Scale and Hoehn and Yahr scale disability scores.⁸⁷

The association between pathological changes seen on OCT and other neurodegenerative and mitochondrial disorders has been highlighted in the past decade. For example, thinning of the pRNFL and mGCIPL has been reported in people with Huntington's disease, Friedreich's ataxia, and spinocerebellar ataxia, with some reports indicating preferential temporal sector pRNFL thinning

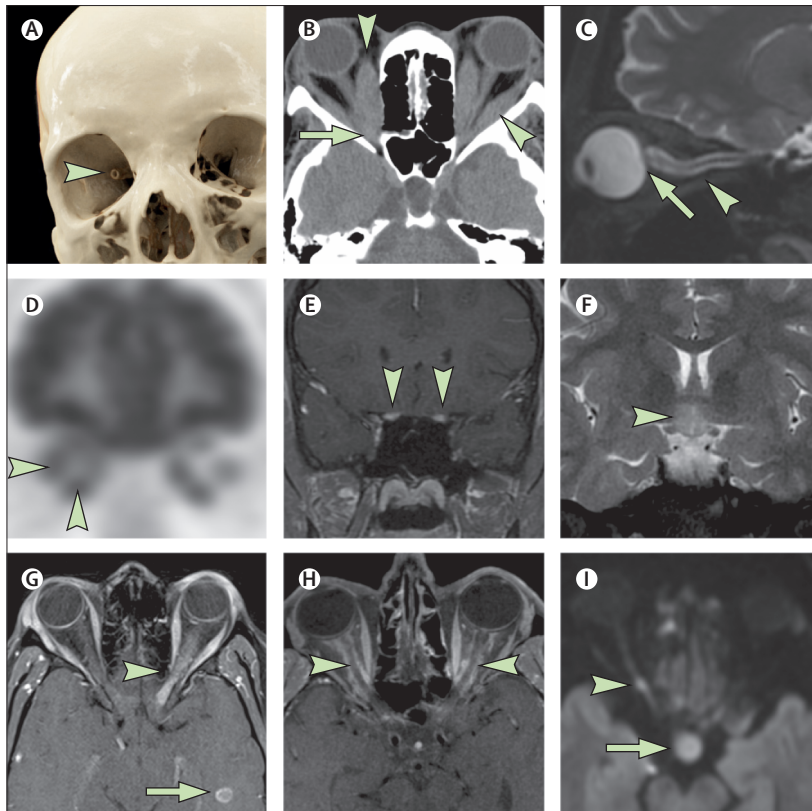


Figure 3: Applications of orbital imaging

(A) Calcified optic nerve sheath meningioma: cinematic volume-rendered reconstruction from a non-contrast CT of the orbits produces a more photorealistic representation with enhanced depth and shape perception. Calcification (arrowhead) from a meningioma circumferentially surrounds the posterior intra-orbital right optic nerve. (B) Thyroid orbitopathy: axial non-contrast CT image through the orbits showing enlargement of bilateral extraocular muscles (arrowheads), most prominently involving the medial rectus muscle. There is crowding of both orbital apices, with prominent loss of fat surrounding the intra-orbital right optic nerve (arrow). (C) Idiopathic intracranial hypertension: curved multiplanar reformation along the axis of the left optic nerve from a volumetric T2-weighted fat-saturated MRI acquisition. There is vertical tortuosity of the optic nerve (arrowhead) and posterior scleral flattening (arrow) from elevated intracranial pressure. (D) Orbital lymphoma: coronal attenuation-corrected PET image from a [^{18}F] fluorodeoxyglucose-PET-CT examination. Multiple bilateral extraocular muscles (arrowheads) show abnormal hypermetabolism due to infiltration by marginal zone B-cell lymphoma. Associated infiltrative intraconal soft tissue (not shown) compressed the right optic nerve within the orbital apex. (E, F) Bilateral optic neuritis from NMO: coronal T1-weighted fat-saturated image (E) from an orbital MRI showing enhancement of the intra-canalicular segments of both optic nerves (arrowheads). Coronal T2-weighted fat-saturated image (F) shows abnormally increased T2 signal within the hypothalamus in the setting of NMO. (G) Optic neuritis from multiple sclerosis: axial T1-weighted fat-saturated image from an orbital MRI showing enhancement of the left optic nerve (arrowhead). An enhancing lesion within the left temporal lobe (arrow) is seen, along with other enhancing and non-enhancing lesions (not shown) in a distribution consistent with multiple sclerosis. (H) Giant cell arteritis: axial T1-weighted fat-saturated image from an orbital MRI showing inflammatory enhancement (arrowheads) along both optic nerves in a patient who presented with bilateral sequential arteritic anterior ischemic optic neuropathy from biopsy-proven giant cell arteritis. (I) Optic nerve infarction: axial diffusion-weighted image through the orbits shows focal diffusion hyperintensity (arrowhead) within the right intra-orbital optic nerve that exhibited corresponding dropout on apparent diffusivity coefficient (not shown) consistent with infarction. This infarction was a complication from endoscopic transsphenoidal surgery for a pituitary adenoma (arrow; partly shown) that resulted in right cavernous internal carotid artery dissection with ophthalmic artery occlusion. NMO=neuromyelitis optica spectrum disorder.

corresponding to the small axonal fibres found in the papillomacular bundle.^{89,90} Furthermore, several studies have highlighted the association between pRNFL and mGCIPL thinning and clinical measures of Huntington's disease severity, including the Unified Huntington's Disease Rating Scale, trinucleotide CAG repeat numbers,

and disease duration.⁸⁹ Preliminary evidence suggests that pRNFL and mGCIPL measurements are inversely correlated with the Scale for the Assessment and Rating of Ataxia scores in patients with Friedreich's ataxia and those with spinocerebellar ataxia.⁹⁰

Use of OCT in retrograde trans-synaptic degeneration from retrogeniculate lesions

Measurement of mGCIPL thickness on OCT is being used increasingly to detect retrograde trans-synaptic degeneration in patients with homonymous hemianopias secondary to post-geniculate lesions involving the intracranial visual pathways. This technique can sometimes substantiate a homonymous hemianopia when visual field testing is unreliable (appendix p 21).⁹¹ Macular thinning, in a pattern corresponding to the visual field defect, has been reported as early as 5 months after the lesion (as opposed to as early as 3–4 weeks in pre-geniculate lesions involving the optic tract and chiasm), whereas evidence of changes in the pRNFL is seen much later, or never develops.⁹¹

Orbital imaging

Cross-sectional imaging modalities, including CT and MRI, directly visualise the entire course of the optic nerves, differentiate various causes of extrinsic compression, distinguish active and chronic processes involving the optic nerves, and identify associated intracranial abnormalities supporting specific diagnoses (figure 3). Advances in scanner technology and image acquisition and reconstruction techniques have enabled accelerated generation of images with increased spatial resolution, providing tissue-specific and physiological information. Computational power facilitates rapid processing and reconstruction of vast amounts of image data. Molecular imaging techniques (eg, PET-CT and PET-MRI hybrid systems) can contemporaneously acquire structural images and metabolic and functional information that help characterise lesions.⁹²

Orbital CT

Modern orbital CT uses multi-detector row helical scanners and can rapidly produce submillimetre slice thicknesses for clear delineation of the bony orbit, optic nerve, and other orbital soft tissue structures. CT optimally shows osseous orbital lesions, intra-lesional calcification, hyperostosis from meningiomas, and orbital apex crowding from thyroid orbitopathy;⁹³ however, resultant optic nerve abnormalities are better assessed by MRI. Administration of intravenous iodinated contrast material increases the conspicuity of orbital inflammation and neovascularity. Advances in detector technology and iterative image reconstruction algorithms have led to improved image quality while decreasing radiation dose. Dual-energy and multi-spectral CT technology make use of differential absorption of various energy x-rays to characterise calcification, haemorrhage, and iodinated contrast. Routine

	Consequence	Potential solutions
Inadequate spatial resolution	Decreased detection of fine details within the orbit, including loss of sensitivity to optic nerve abnormalities	Decrease FOV tailored to orbit; decrease slice thickness; volumetric isotropic sequences (eg, SPACE); increased channel density head coil or dedicated surface coils (for superficial orbital structures); increased magnetic field strength (3 T or 7 T)
Incomplete fat-saturation using conventional frequency-selective fat saturation	False-negative for optic nerve sheath or other extrinsic orbital masses due to non-suppressed adjacent fat; false positive for atypical orbital lesion due to localised incomplete fat saturation	Reduced FOV approaches; short tau inversion recovery; Dixon method (chemical shift-based) fat-suppression for T1-weighted and T2-weighted images; VIBE or radial VIBE for fat-suppression on T1-weighted images
Susceptibility artifact from adjacent metal (eg, dental amalgam or postsurgical changes)	False-negative for abnormal optic nerve signal or enhancement due to obscuration by artifact; false positive for atypical optic nerve signal or enhancement due to field distortion and propagation of artifact onto the optic nerve	Reduced FOV approaches; metal suppression sequences
Artifact from eye motion or head motion	Motion artifacts with obscuration of fine detail; if severe, completely non-diagnostic imaging	Ultrafast acquisitions using acceleration techniques (parallel imaging, simultaneous multislice acquisition) and artificial intelligence-based image reconstructions; real-time head motion correction; head immobilisation and external eye gaze fixation
Difficulty detecting atypical optic nerve T2 signal or enhancement on T1	False-negatives for optic nerve lesions	Double inversion recovery; turbo-spin-echo based three-dimensional T1 sequences (eg, T1-mVISTA)
Suboptimal DWI	Image distortion and obscuration of diffusion abnormalities such as optic nerve infarction	Reduced FOV, non-echo-planar imaging-based DWI approaches using fast-spin-echo or readout segmented acquisitions (eg, RESOLVE)
Low specificity of MRI for detected orbital masses	Need for additional imaging or tissue sampling	MRI-PET hybrid systems with simultaneous acquisitions using [¹⁸ F] fluorodeoxyglucose or novel PET radiotracers

DWI=diffusion-weighted imaging. FOV=field of view. mVISTA=volume isotropic turbo spin echo acquisition. RESOLVE=readout segmentation of long variable echo trains. SPACE=sampling perfection with application optimised contrasts using different flip angle evolution. VIBE=volumetric interpolated breath-hold examination.

Table 2: Technical challenges, consequences, and possible solutions for orbital MRI

multiplanar reconstruction and volume rendering allows for myriad approaches to visualisation of the orbit.

Orbital MRI

With its superior contrast resolution and multiple image contrast mechanisms, MRI is preferable to CT for most optic nerve applications (table 2).⁶ The optimal technique entails a dedicated protocol tailored to the orbits, reducing field of view and slice thickness to increase spatial resolution, adding coronal plane sequences, and using fat-saturation to improve depiction of orbital structures and lesion conspicuity. Advances in MRI scanner technology have yielded superior imaging capabilities because of improved magnetic field homogeneity and augmented gradient coil performance. Units with increased magnetic field strength (eg, 3 T and new 7 T models) provide intrinsically enhanced signal-to-noise, which can be applied for high-resolution imaging or accelerated acquisition. Innovative radio-frequency coil technology in head coils has largely supplanted the need for surface coils; however, dedicated orbital coils could be applied for the imaging of selected orbital anatomical structures.^{6,94} Novel sequences speed up image acquisition through under-sampling, in-slice acceleration using parallel imaging, and multi-band simultaneous slice acquisition, which mitigates eye movement and head motion artifacts. In the future, real-time motion-correction and artificial intelligence-based image reconstruction could advance the

boundaries of fast, high-resolution, motion-insensitive orbital MRI.

Effective orbital MRI requires dependable fat-saturation to accentuate atypical T2 signal within the optic nerve and enhancement on T1-weighted images.^{6,95} Bone-air interfaces with the paranasal sinuses, and susceptibility artifacts from dental amalgam and postsurgical metal, can prevent effective fat-saturation, thereby diminishing diagnostic confidence and generating artifacts, leading to false-positive findings. Innovative techniques for fat-saturation (eg, Dixon method to achieve uniform fat suppression) improve on conventional frequency-selective and short tau inversion recovery approaches, and metal-artifact suppression techniques reduce susceptibility artifacts from local metal.⁶ Reduced field of view methods selectively image a region, reducing propagation of extrinsic artifacts. Volumetric high-resolution sequences enable reconstruction in any plane, including along the optic nerve, making these sequences ideally suited for detection of optic nerve findings, such as tortuosity and nerve sheath distension in idiopathic intracranial hypertension.^{6,95}

Intravenous administration of a gadolinium-based contrast agent improves conspicuity of inflammation, neovascularity, and other processes, exploiting the absence of enhancement seen with healthy optic nerves. High-resolution volumetric fat-saturated post-contrast T1-weighted imaging, using techniques such as

Panel 4: The power of multimodality imaging of the optic nerve

Up to 10 years ago, the use of multiple methods of testing for the optic nerve was mostly restricted to tests of function (eg, visual acuity, colour vision, and visual field testing) combined with structural fundusoscopic appearance of the optic nerve head and standard imaging by CT or MRI of the orbital and intracranial nerve.¹ The ocular imaging techniques highlighted in this review and in the two other reviews in *The Lancet Neurology Series* on optic neuropathies^{2,3} not only provide improved structural imaging in such exquisite detail so as to allow exact localisation of pathology, but also can offer clues to the underlying pathophysiology and even pinpoint the chronicity of the pathological process. The combination of these now routinely used optic nerve imaging modalities has become standard practice because each tool provides objective, varied, and yet complementary information.⁴⁻⁶

For example, in a patient complaining of visual loss immediately after a traumatic brain injury, findings of an optic neuropathy should raise concern for an acute traumatic optic neuropathy. However, the discovery of thin mGCIPL on OCT within a few days of the trauma affirms the chronicity of the optic neuropathy and rules out the recent trauma as the cause of visual loss. Multimodality optic nerve imaging is especially useful when evaluating patients with possible optic disc oedema (appendix pp 16, 19, 22). Ocular fundus photographs are essential to document the optic nerve appearance, including the cup-to-disc ratio and the colour of the neural rim, as well as the presence of vessel obscuration and haemorrhages or exudates. OCT with enhanced depth imaging complements simple photography by providing direct imaging of deeper optic nerve

structures and showing optic nerve drusen or peripapillary hyper-reflective ovoid mass-like structures, findings that might suggest pseudo-disc oedema rather than true disc oedema. Orbital ultrasound is also often used in this setting to show calcified optic disc drusen. OCT of the pRNFL can serve as a reference for future comparison, whereas serial imaging will allow for close monitoring of the pRNFL thickness in response to the treatment of patients with papilloedema. The combination of pRNFL OCT with mGCIPL provides specific information on the regression of disc oedema and allows for differentiation between the development of secondary optic atrophy (in which case the pRNFL thickness normalises but the mGCIPL thins, which is consistent with retinal ganglion cell atrophy) and a successful treatment (in which case the pRNFL thickness normalises and the mGCIPL remains within a normal range). Objective evaluation of disease activity with various OCT techniques is routinely used to complement subjective tests of visual function in patients with compressive optic neuropathies (figure 2) or those with optic neuritis associated with various neurological inflammatory disorders.

Hence, in addition to providing useful information regarding localisation of the pathological process, differential diagnosis of the underlying causes, and characterisation of disease characteristics, integrative multimodality imaging of the optic nerve allows for evaluation of disease activity and monitoring of response to treatment.

mGCIPL=macular ganglion cell-inner plexiform layer. OCT=optical coherence tomography. pRNFL=papillary retinal nerve fibre layer.

volumetric interpolated breath-holding examination, can facilitate optimal visualisation of optic nerve enhancement.^{6,95} When MRI is obtained within 3 weeks of symptom onset, enhancement of the optic nerve is reportedly seen in at least 90% of instances of optic neuritis.⁹⁶ However, findings of enhancement are likely to be dependent on the MRI techniques and protocols used. Novel contrast mechanisms, including double inversion recovery, can increase sensitivity for atypical optic nerve signal. Contrast-enhancement, enlargement, and high T2 signal oedema are typically seen in cases of acute optic neuritis, and decreased calibre and high T2 signal are seen in the chronic setting. However, these findings are not cause specific. Associated brain findings characteristic of demyelination in multiple sclerosis or NMOSD, enhancing brain or leptomeningeal lesions in sarcoidosis, and other associated intracranial findings can be very valuable for differential diagnosis. Optic nerve sheath enhancement and orbital fat stranding are common in people with MOGAD, in addition to optic nerve enhancement.² Mass-like enhancement from NMOSD can be confused during imaging for other intrinsic lesions of the optic nerve, such as optic pathway gliomas.⁹⁷ Compared with other imaging techniques,

MRI excels at detection of extrinsic or compressive lesions, including optic nerve sheath meningiomas that characteristically show a so-called tram track enhancement along the nerve. Occasionally, this pattern can be confused with inflammatory optic perineuritis or metastatic disease.⁹⁸ Although sellar lesions, such as pituitary macroadenomas, typically cause chiasmatic compression, optic nerves might be compressed in the setting of a post-fixed chiasm.

MRI using diffusion-weighted imaging (DWI) detects restriction of free water movement within and along cells and is integral to brain imaging for detection of ischaemia. Optic nerve ischaemia can be detected by DWI in the setting of arteritic ischaemic optic neuropathies and perioperative posterior ischaemic optic neuropathies. Associated findings of brain infarction and temporal artery inflammation can also be detected by MRI. Standard DWI, using single-shot echo planar imaging, is challenging in the orbits due to numerous artifacts. Novel DWI approaches, including reduced field of view, fast-spin-echo-based techniques, and readout segmented techniques such as REadout Segmentation Of Long Variable Echo trains (RESOLVE) offer increased resolution, decreased distortion from adjacent bony

structures, and improved lesion conspicuity compared with other DWI approaches.⁹⁹ Increasing diffusion-encoding directions permits diffusion tensor imaging and the potential for relatively artifact-free tractography of the optic nerve in clinically relevant timeframes.¹⁰⁰ Other techniques (eg, susceptibility-weighted imaging) have been applied in the orbits, such as for the detection of calcifications within optic nerve sheath meningiomas.

Conclusions and future directions

Techniques for optic nerve imaging have improved greatly during the past 20 years. Multimodality imaging has become standard practice because each different technology provides varied and complementary information (panel 4). Increased widespread use of non-mydratic fundus photography offers unique opportunities for neurologists to easily examine the optic nerve appearance and facilitate early diagnosis of neurological conditions, with the help of existing teleneurology and teleophthalmology networks.¹⁸ With the increase in artificial intelligence algorithms in medical imaging, we are beginning to see tools that can be used as diagnostic aids, providing assistance for interpretation of optic nerve appearance and diagnosis of papilloedema in emergency settings.³⁸

The latest generation of OCTs allows for routine, rapid, and non-invasive detection of subtle changes in pRNFL, mGCIPL, deep optic nerve structures, and vasculature in patients with all causes of optic neuropathies and in those with various neurological disorders, providing objective markers useful for the diagnosis of optic neuropathies and for patient monitoring and measuring outcomes, particularly in clinical trials.^{2,4,5} Advanced OCT technology, including ultrahigh-resolution spectral-domain OCT systems for imaging the fine structural details of the retina, and ultrahigh-speed swept-source systems for OCT-A of the retinal vasculature, will allow for reproducible measurement of preclinical changes, which are important for understanding pathogenesis and developing early therapies for various neurological disorders. Multimodal optic nerve imaging, including EDI-OCT, has also resulted in the description of new pathological findings such as PHOMS.¹⁰¹

New orbital imaging studies using readily available radiological technology allow for direct high-resolution imaging of the optic nerve, greatly facilitating the diagnosis of various optic neuropathies.⁶ Although most of the existing ocular imaging technologies are generally used for assessing anatomical or structural changes, the modification of existing technology makes functional imaging of metabolic activity possible.¹⁰² The combination of automated optic nerve analysis with AI-DLS and large scale genomic analysis¹⁰³ is another example of the many potential applications of optic nerve imaging. Incorporation of novel optic nerve imaging modalities into the neurological examination will not only improve diagnostic capabilities but also facilitate objective

Search strategy and selection criteria

We searched PubMed, the Cochrane library, MEDLINE, and Embase between Jan 1, 2016, and August 1, 2022, using the search terms “optic neuropathy”, “optic nerve imaging”, “retinal fundus photography”, “smartphone imaging”, “orbital ultrasound”, “optical coherence tomography”, “optical coherence tomography angiography”, “orbital MRI”, “orbital computed tomography”, “orbital imaging”, “deep learning”, and “artificial intelligence”. Key older references were selected by the authors if judged appropriate to the review. We also reviewed the reference lists of each selected article to identify other relevant publications. We restricted our search to publications in the English language. The final reference list was generated on the basis of relevance to the topics covered in this review and the quality of each reviewed publication.

monitoring and provide outcome measures useful at both the patient level and in clinical trials.

Contributors

All authors contributed equally to the preparation of the manuscript, including review and selection of relevant publications, collection of data, preparation of the first draft, review of the final draft, and preparation of the tables and figures.

Declaration of interests

CL is a consultant for Allergan France, Novartis France, and Horus Pharma. VB is a consultant for GenSight Biologics and Neurophoenix, and receives research support from GenSight Biologics and Santhera/Chiesi. NJN is a consultant for GenSight Biologics, Santhera/Chiesi, Stealth Biotherapeutics, and Neurophoenix; receives research support from GenSight Biologics and Santhera/Chiesi; is a participant in educational webinars sponsored by WebMD-Global Medscape and First Class; and is a medical-legal consultant in matters not related to this work. All other authors declare no competing interests.

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References

- 1 Bioussé V, Newman NJ. Diagnosis and clinical features of common optic neuropathies. *Lancet Neurol* 2016; **15**: 1355–67.
- 2 Bennett JL, Costello F, Chen JJ, et al. Optic neuritis and autoimmune optic neuropathies: advances in diagnosis and treatment. *Lancet Neurol* 2022; published online Sept 22. [https://doi.org/10.1016/S1474-4422\(22\)00187-9](https://doi.org/10.1016/S1474-4422(22)00187-9).
- 3 Newman NJ, Yu-Wai-Man P, Bioussé V, Carelli V. Understanding the molecular basis and pathogenesis of hereditary optic neuropathies: towards improved diagnosis and management. *Lancet Neurol* 2022; published online Sept 22. [https://doi.org/10.1016/S1474-4422\(22\)00174-0](https://doi.org/10.1016/S1474-4422(22)00174-0).
- 4 Minakaran N, de Carvalho ER, Petzold A, Wong SH. Optical coherence tomography (OCT) in neuro-ophthalmology. *Eye (Lond)* 2021; **35**: 17–32.
- 5 Pujari A, Bhaskaran K, Sharma P, et al. Optical coherence tomography angiography in neuro-ophthalmology: current clinical role and future perspectives. *Surv Ophthalmol* 2021; **66**: 471–81.
- 6 Chow LS, Paley MNJ. Recent advances on optic nerve magnetic resonance imaging and post-processing. *Magn Reson Imaging* 2021; **79**: 76–84.
- 7 Hatcher-Martin JM, Busis NA, Cohen BH, et al. American academy of neurology telehealth position statement. *Neurology* 2021; **97**: 334–39.

- 8 Woodward MA, Bavinger JC, Amin S, et al. Telemedicine for ophthalmic consultation services: use of a portable device and layering information for graders. *J Telemed Telecare* 2017; **23**: 365–70.
- 9 Rajalakshmi R, Prathiba V, Arulmalar S, Usha M. Review of retinal cameras for global coverage of diabetic retinopathy screening. *Eye (Lond)* 2021; **35**: 162–72.
- 10 Vasseneix C, Bruce BB, Bidot S, Newman NJ, Biousse V. Nonmydriatic fundus photography in patients with acute vision loss. *Telemed J E Health* 2019; **25**: 911–16.
- 11 De Arrigunaga S, Aziz K, Lorch AC, Friedman DS, Armstrong GW. A review of ophthalmic telemedicine for emergency department settings. *Semin Ophthalmol* 2022; **37**: 83–90.
- 12 Ko MW, Busis NA. Tele-neuro-ophthalmology: vision for 20/20 and beyond. *J Neuroophthalmol* 2020; **40**: 378–84.
- 13 Liu YA, Ko MW, Moss HE. Telemedicine for neuro-ophthalmology: challenges and opportunities. *Curr Opin Neurol* 2021; **34**: 61–66.
- 14 Calix R, Grossman SN, Rasool N, et al. Practical approach to the tele-neuro-ophthalmology and neuro-otology visits: instructional videos. *J Neuroophthalmol* 2021; **41**: 10–12.
- 15 Nagra M, Konstantopoulos A, Pardhan S. Smartphone ophthalmoscopy: a potential way forward for non-ophthalmology medics in the COVID-19 era. *BMJ Innov* 2021; **7**: 237–39.
- 16 Shih KC, Chau CYC, Chan JCH, Wong JKW, Lai JSM. Does the COVID-19 pandemic spell the end for the direct ophthalmoscope? *Ophthalmol Ther* 2020; **9**: 689–92.
- 17 Mackay DD, Garza PS, Bruce BB, Newman NJ, Biousse V. The demise of direct ophthalmoscopy: a modern clinical challenge. *Neurol Clin Pract* 2015; **5**: 150–57.
- 18 Biousse V, Bruce BB, Newman NJ. Ophthalmoscopy in the 21st century: the 2017 H. Houston Merritt Lecture. *Neurology* 2018; **90**: 167–75.
- 19 Zafar S, Cardenas YM, Leishangthem L, Yaddanapudi S. Opinion and special articles: amateur fundus photography with various new devices: our experience as neurology residents. *Neurology* 2018; **90**: 897–901.
- 20 Dunn HP, Kang CJ, Marks S, et al. Perceived usefulness and ease of use of funduscopy by medical students: a randomised crossover trial of six technologies (eFOCUS 1). *BMC Med Educ* 2021; **21**: 41.
- 21 Panwar N, Huang P, Lee J, et al. Fundus photography in the 21st century—a review of recent technological advances and their implications for worldwide healthcare. *Telemed J E Health* 2016; **22**: 198–208.
- 22 Kelly LP, Garza PS, Bruce BB, Graubart EB, Newman NJ, Biousse V. Teaching ophthalmoscopy to medical students (the TOTeMS study). *Am J Ophthalmol* 2013; **156**: 1056–61.
- 23 Toffoli D, Bruce BB, Lamirel C, Henderson AD, Newman NJ, Biousse V. Feasibility and quality of nonmydriatic fundus photography in children. *J AAPOS* 2011; **15**: 567–72.
- 24 Ivan Y, Ramgopal S, Cardenas-Villa M, et al. Feasibility of the digital retinography system camera in the pediatric emergency department. *Pediatr Emerg Care* 2018; **34**: 488–91.
- 25 Kim Y, Chao DL. Comparison of smartphone ophthalmoscopy vs conventional direct ophthalmoscopy as a teaching tool for medical students: the COSMOS study. *Clin Ophthalmol* 2019; **13**: 391–401.
- 26 Bursztyl L, Woodward MA, Cornblath WT, et al. Accuracy and reliability of a handheld, nonmydriatic fundus camera for the remote detection of optic disc edema. *Telemed J E Health* 2018; **24**: 344–50.
- 27 Sachdeva V, Vasseneix C, Hage R, et al. Optic nerve head edema among patients presenting to the emergency department. *Neurology* 2018; **90**: e373–79.
- 28 Bruce BB, Biousse V, Newman NJ. Nonmydriatic ocular fundus photography in neurologic emergencies. *JAMA Neurol* 2015; **72**: 455–59.
- 29 Sharma RA, Garza PS, Biousse V, Samuels OB, Newman NJ, Bruce BB. Ocular fundus abnormalities in acute subarachnoid hemorrhage: the FOTO-ICU study. *Neurosurgery* 2021; **88**: 278–84.
- 30 Bruce BB, Lamirel C, Wright DW, et al. Nonmydriatic ocular fundus photography in the emergency department. *N Engl J Med* 2011; **364**: 387–89.
- 31 Bruce BB, Thulasi P, Fraser CL, et al. Diagnostic accuracy and use of nonmydriatic ocular fundus photography by emergency physicians: phase II of the FOTO-ED study. *Ann Emerg Med* 2013; **62**: 28–33.
- 32 Dunn HP, Teo KZ, Smyth JW, et al. Using non-mydratic fundus photography to detect fundus pathology in Australian metropolitan emergency departments: a prospective prevalence and diagnostic accuracy study. *Emerg Med Australas* 2021; **33**: 302–09.
- 33 Irani NK, Bidot S, Peragallo JH, Esper GJ, Newman NJ, Biousse V. Feasibility of a nonmydriatic ocular fundus camera in an outpatient neurology clinic. *Neurologist* 2020; **25**: 19–23.
- 34 Bruce BB, Bidot S, Hage R, et al. Fundus photography vs. ophthalmoscopy outcomes in the emergency department (FOTO-ED) phase III: web-based, in-service training of emergency providers. *Neuroophthalmology* 2018; **42**: 269–74.
- 35 John White. Criminal trial of optom. *Optometry Today*. 2016. <https://www.aop.org.uk/ot/professional-support/clinical-and-regulatory/2016/07/14/criminal-trial-of-optometrist-begins> (accessed March 28, 2022).
- 36 Li B, Chen H, Zhang B, et al. Development and evaluation of a deep learning model for the detection of multiple fundus diseases based on colour fundus photography. *Br J Ophthalmol* 2021; published online March 30. <https://doi.org/10.1136/bjophthalmol-2020-316290>.
- 37 Channa R, Wolf R, Abràmoff MD. Autonomous artificial intelligence in diabetic retinopathy: from algorithm to clinical application. *J Diabetes Sci Technol* 2021; **15**: 695–98.
- 38 Milea D, Najjar RP, Zhuho J, et al. Artificial intelligence to detect papilledema from ocular fundus photographs. *N Engl J Med* 2020; **382**: 1687–95.
- 39 Vasseneix C, Najjar RP, Xu X, et al. Accuracy of a deep learning system for classification of papilledema severity on ocular fundus photographs. *Neurology* 2021; **97**: e369–77.
- 40 Biousse V, Newman NJ, Najjar RP, et al. Optic disc classification by deep learning versus expert neuro-ophthalmologists. *Ann Neurol* 2020; **88**: 785–95.
- 41 Stunkel L, Sharma RA, Mackay DD, et al. Patient harm due to diagnostic error of neuro-ophthalmologic conditions. *Ophthalmology* 2021; **128**: 1356–62.
- 42 Yan Y, Liao YJ. Updates on ophthalmic imaging features of optic disc drusen, papilledema, and optic disc edema. *Curr Opin Neurol* 2021; **34**: 108–15.
- 43 Skidmore C, Saurey T, Ferre RM, Rodriguez-Brizuela R, Spaulding J, Lundgreen Mason N. A narrative review of common uses of ophthalmic ultrasound in emergency medicine. *J Emerg Med* 2021; **60**: 80–89.
- 44 Yildizdas D, Aslan N. Is ocular sonography a reliable method for the assessment of elevated intracranial pressure in children? *J Pediatr Intensive Care* 2021; **10**: 14–22.
- 45 Gandhi K, Shyy W, Knight S, Teismann N. Point-of-care ultrasound for the evaluation of non-traumatic visual disturbances in the emergency department: the VIGMO protocol. *Am J Emerg Med* 2019; **37**: 1547–53.
- 46 Lochner P, Leone MA, Coppo L, et al. B-mode transorbital ultrasonography for the diagnosis of acute optic neuritis. A systematic review. *Clin Neurophysiol* 2016; **127**: 803–09.
- 47 Petzold A, Balcer LJ, Calabresi PA, et al. Retinal layer segmentation in multiple sclerosis: a systematic review and meta-analysis. *Lancet Neurol* 2017; **16**: 797–812.
- 48 Ulusoy MO, Horasanlı B, Işık-Ulusoy S. Optical coherence tomography angiography findings of multiple sclerosis with or without optic neuritis. *Neurol Res* 2020; **42**: 319–26.
- 49 Murphy OC, Kwakyi O, Ifikhar M, et al. Alterations in the retinal vasculature occur in multiple sclerosis and exhibit novel correlations with disability and visual function measures. *Mult Scler* 2020; **26**: 815–28.
- 50 Murphy OC, Kalaitzidis G, Vasileiou E, et al. Optical coherence tomography and optical coherence tomography angiography findings after optic neuritis in multiple sclerosis. *Front Neurol* 2020; **11**: 618879.
- 51 Lampert EJ, Andorra M, Torres-Torres R, et al. Color vision impairment in multiple sclerosis points to retinal ganglion cell damage. *J Neurol* 2015; **262**: 2491–97.

- 52 Triplett JD, Yiannikas C, Barnett MH, et al. Pathophysiological basis of low contrast visual acuity loss in multiple sclerosis. *Ann Clin Transl Neurol* 2018; **5**: 1505–12.
- 53 Knier B, Leppenietter G, Wetzlmair C, et al. Association of retinal architecture, intrathecal immunity, and clinical course in multiple sclerosis. *JAMA Neurol* 2017; **74**: 847–56.
- 54 Koraysha NA, Kishk N, Hassan A, et al. Evaluating optic nerve diameter as a possible biomarker for disability in patients with multiple sclerosis. *Neuropsychiatr Dis Treat* 2019; **15**: 2571–78.
- 55 Klistorner A, Graham EC, Yiannikas C, et al. Progression of retinal ganglion cell loss in multiple sclerosis is associated with new lesions in the optic radiations. *Eur J Neurol* 2017; **24**: 1392–98.
- 56 Petracca M, Cordano C, Cellerino M, et al. Retinal degeneration in primary-progressive multiple sclerosis: a role for cortical lesions? *Mult Scler* 2017; **23**: 43–50.
- 57 Saidha S, Al-Louzi O, Ratchford JN, et al. Optical coherence tomography reflects brain atrophy in multiple sclerosis: a four-year study. *Ann Neurol* 2015; **78**: 801–13.
- 58 Lambe J, Fitzgerald KC, Murphy OC, et al. Association of spectral-domain OCT with long-term disability worsening in multiple sclerosis. *Neurology* 2021; **96**: e2058–69.
- 59 Bsteh G, Hegen H, Teuchner B, et al. Peripapillary retinal nerve fibre layer as measured by optical coherence tomography is a prognostic biomarker not only for physical but also for cognitive disability progression in multiple sclerosis. *Mult Scler* 2019; **25**: 196–203.
- 60 Gabilondo I, Martínez-Lapiscina EH, Fraga-Pumar E, et al. Dynamics of retinal injury after acute optic neuritis. *Ann Neurol* 2015; **77**: 517–28.
- 61 Kupersmith MJ, Garvin MK, Wang JK, Durbin M, Kardon R. Retinal ganglion cell layer thinning within one month of presentation for optic neuritis. *Mult Scler* 2016; **22**: 641–48.
- 62 Filippatou AG, Mukharesh L, Saidha S, Calabresi PA, Sotirchos ES. AQP4-IgG and MOG-IgG related optic neuritis-prevalence, optical coherence tomography findings, and visual outcomes: a systematic review and meta-analysis. *Front Neurol* 2020; **11**: 540156.
- 63 Naismith RT, Tutlam NT, Xu J, et al. Optical coherence tomography differs in neuromyelitis optica compared with multiple sclerosis. *Neurology* 2009; **72**: 1077–82.
- 64 Aly L, Strauß EM, Feucht N, et al. Optical coherence tomography angiography indicates subclinical retinal disease in neuromyelitis optica spectrum disorders. *Mult Scler* 2022; **28**: 522–31.
- 65 Huang-Link Y, Eleftheriou A, Yang G, et al. Optical coherence tomography represents a sensitive and reliable tool for routine monitoring of idiopathic intracranial hypertension with and without papilledema. *Eur J Neurol* 2019; **26**: 808–e57.
- 66 Malmqvist L, Bursztyn L, Costello F, et al. The optic disc drusen studies consortium recommendations for diagnosis of optic disc drusen using optical coherence tomography. *J Neuroophthalmol* 2018; **38**: 299–307.
- 67 Vijay V, Mollan SP, Mitchell JL, et al. Using optical coherence tomography as a surrogate of measurements of intracranial pressure in idiopathic intracranial hypertension. *JAMA Ophthalmol* 2020; **138**: 1264–71.
- 68 Athappilly G, García-Basterra I, Machado-Miller F, Hedges TR, Mendoza-Santisteban C, Vuong L. Ganglion cell complex analysis as a potential indicator of early neuronal loss in idiopathic intracranial hypertension. *Neuroophthalmology* 2018; **43**: 10–17.
- 69 Sibony PA, Kupersmith MJ, Feldon SE, Wang J-K, Garvin M, OCT Substudy Group. Retinal and choroidal folds in papilledema. *Invest Ophthalmol Vis Sci* 2015; **56**: 5670–80.
- 70 Kupersmith MJ, Sibony PA, Feldon SE, Wang JK, Garvin M, Kardon R. The effect of treatment of idiopathic intracranial hypertension on prevalence of retinal and choroidal folds. *Am J Ophthalmol* 2017; **176**: 77–86.
- 71 Malhotra K, Patel MD, Shirazi Z, Moss HE. Association between peripapillary bruch's membrane shape and intracranial pressure: effect of image acquisition pattern and image analysis method, a preliminary study. *Front Neurol* 2018; **9**: 1137.
- 72 Danesh-Meyer HV, Yoon JJ, Lawlor M, Savino PJ. Visual loss and recovery in chiasmal compression. *Prog Retin Eye Res* 2019; **73**: 100765.
- 73 Danesh-Meyer HV, Carroll SC, Foroosan R, et al. Relationship between retinal nerve fiber layer and visual field sensitivity as measured by optical coherence tomography in chiasmal compression. *Invest Ophthalmol Vis Sci* 2006; **47**: 4827–35.
- 74 Tieger MG, Hedges TR 3rd, Ho J, et al. Ganglion cell complex loss in chiasmal compression by brain tumors. *J Neuroophthalmol* 2017; **37**: 7–12.
- 75 Blanch RJ, Micieli JA, Oyesiku NM, Newman NJ, Biousse V. Optical coherence tomography retinal ganglion cell complex analysis for the detection of early chiasmal compression. *Pituitary* 2018; **21**: 515–23.
- 76 Wang X, Chou Y, Zhu H, et al. Retinal microvascular alterations detected by optical coherence tomography angiography in nonfunctioning pituitary adenomas. *Transl Vis Sci Technol* 2022; **11**: 5.
- 77 Wang MTM, King J, Symons RCA, et al. Temporal patterns of visual recovery following pituitary tumor resection: a prospective cohort study. *J Clin Neurosci* 2021; **86**: 252–59.
- 78 Wang MTM, King J, Symons RCA, et al. Prognostic utility of optical coherence tomography for long-term visual recovery following pituitary tumor surgery. *Am J Ophthalmol* 2020; **218**: 247–54.
- 79 Lee J, Kim SW, Kim DW, et al. Predictive model for recovery of visual field after surgery of pituitary adenoma. *J Neurooncol* 2016; **130**: 155–64.
- 80 Akbari M, Abdi P, Fard MA, et al. Retinal ganglion cell loss precedes retinal nerve fiber thinning in nonarteritic anterior ischemic optic neuropathy. *J Neuroophthalmol* 2016; **36**: 141–46.
- 81 MacIntosh PW, Kumar SV, Saravanan VR, Shah VM. Acute changes in ganglion cell layer thickness in ischemic optic neuropathy compared to optic neuritis using optical coherence tomography. *Int J Ophthalmol* 2020; **13**: 120–23.
- 82 Xiao H, Liu X, Lian P, Liao LL, Zhong YM. Different damage patterns of retinal nerve fiber layer and ganglion cell-inner plexiform layer between early glaucoma and non-glaucomatous optic neuropathy. *Int J Ophthalmol* 2020; **13**: 893–901.
- 83 Darvizeh F, Asanad S, Falavarjani KG, et al. Choroidal thickness and the retinal ganglion cell complex in chronic Leber's hereditary optic neuropathy: a prospective study using swept-source optical coherence tomography. *Eye (Lond)* 2020; **34**: 1624–30.
- 84 Chan VTT, Sun Z, Tang S, et al. Spectral-domain OCT measurements in alzheimer's disease: a systematic review and meta-analysis. *Ophthalmology* 2019; **126**: 497–510.
- 85 Rifai OM, McGrory S, Robbins CB, et al. The application of optical coherence tomography angiography in Alzheimer's disease: a systematic review. *Alzheimers Dement (Amst)* 2021; **13**: e12149.
- 86 Mutlu U, Colijn JM, Ikram MA, et al. Association of retinal neurodegeneration on optical coherence tomography with dementia: a population-based study. *JAMA Neurol* 2018; **75**: 1256–63.
- 87 Chrysou A, Jansson NM, van Laar T. Retinal layers in Parkinson's disease: a meta-analysis of spectral-domain optical coherence tomography studies. *Parkinsonism Relat Disord* 2019; **64**: 40–49.
- 88 Kwapong WR, Ye H, Peng C, et al. Retinal microvascular impairment in the early stages of parkinson's disease. *Invest Ophthalmol Vis Sci* 2018; **59**: 4115–22.
- 89 Kersten HM, Danesh-Meyer HV, Kilfoyle DH, Roxburgh RH. Optical coherence tomography findings in Huntington's disease: a potential biomarker of disease progression. *J Neurol* 2015; **262**: 2457–65.
- 90 Rojas P, Ramírez AI, Hoz R, et al. Ocular involvement in friedreich ataxia patients and its relationship with neurological disability, a follow-up study. *Diagnostics (Basel)* 2020; **10**: 75.
- 91 Meyer BI, Chen BS, Biousse V, Newman NJ. Homonymous thinning on macular optical coherence tomography indicating retrograde trans-synaptic degeneration from occipital infarctions. *Taiwan J Ophthalmol* 2021; **11**: 89–92.
- 92 Kalemaki MS, Karantanis AH, Exarchos D, et al. PET/CT and PET/MRI in ophthalmic oncology (Review). *Int J Oncol* 2020; **56**: 417–29.
- 93 Siakallis LC, Uddin JM, Miszkil KA. Imaging investigation of thyroid eye disease. *Ophthalmic Plast Reconstr Surg* 2018; **34** (4S suppl 1): S41–S51.

- 94 Dobbs NW, Budak MJ, White RD, Zealley IA. MR-Eye: high-resolution microscopy coil MRI for the assessment of the orbit and periorbital structures, part 1: technique and anatomy. *AJNR Am J Neuroradiol* 2020; **41**: 947–50.
- 95 Hoch MJ, Bruno MT, Shepherd TM. Advanced MRI of the optic nerve. *J Neuroophthalmol* 2017; **37**: 187–96.
- 96 Kupersmith MJ, Alban T, Zeiffer B, Lefton D. Contrast-enhanced MRI in acute optic neuritis: relationship to visual performance. *Brain* 2002; **125**: 812–22.
- 97 Clarke L, Arnett S, Lilley K, Liao J, Bhuta S, Broadley SA. Magnetic resonance imaging in neuromyelitis optica spectrum disorder. *Clin Exp Immunol* 2021; **206**: 251–65.
- 98 Ferreira TA, Saraiva P, Genders SW, Buchem MV, Luyten GPM, Beenakker JWCT. CT and MR imaging of orbital inflammation. *Neuroradiology* 2018; **60**: 1253–66.
- 99 Mallon D, Dixon L, Campion T, et al. Beyond the brain: extra-axial pathology on diffusion weighted imaging in neuroimaging. *J Neurol Sci* 2020; **415**: 116900.
- 100 Jacquesson T, Frindel C, Kocevar G, et al. Overcoming challenges of cranial nerve tractography: a targeted review. *Neurosurgery* 2019; **84**: 313–25.
- 101 Fraser JA, Sibony PA, Petzold A, Thaung C, Hamann S, ODDS Consortium. Peripapillary Hyper-reflective Ovoid Mass-like Structure (PHOMS): an optical coherence tomography marker of axoplasmic stasis in the optic nerve head. *J Neuroophthalmol* 2021; **41**: 431–41.
- 102 Gao S, Li Y, Bissig D, et al. Functional regulation of an outer retina hyporeflexive band on optical coherence tomography images. *Sci Rep* 2021; **11**: 10260.
- 103 Han X, Steven K, Qassim A, et al. Automated AI labeling of optic nerve head enables insights into cross-ancestry glaucoma risk and genetic discovery in >280,000 images from UKB and CLSA. *Am J Hum Genet* 2021; **108**: 1204–16.

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